

Classifying Air Quality Index Levels

Based on Pollutant Concentrations in Philippine Urban Environments

By

CSELEC2C Group 4

Baron • Duran • Eva • Rodelas

BACKGROUND AND MOTIVATION

- Air pollution affects public health, mobility, and urban planning.
- AQI transforms pollutant readings into severity levels that are easier to communicate.
- Philippine cities are expanding sensor-based and data-driven monitoring initiatives.
- Machine learning can support analytical interpretation of available air quality records.

ANALYTICAL FOCUS

Monitoring Support

Model Comparison

Interpretability

PROBLEM STATEMENT

CHALLENGE

- AQI labels are ordinal severity classes, not continuous AQI values.
- Lower AQI classes dominate the dataset; severe classes are rare.
- Accuracy alone can overstate performance on imbalanced classes.

RESEARCH QUESTION

Which machine learning model best classifies AQI severity levels from pollutant concentration, temporal, and city-based features?

RESEARCH OBJECTIVES

- 1 BUILD A REPRODUCIBLE PIPELINE**
Merge monthly 2025 CSV files, clean labels, prepare features, and split data.
- 2 COMPARE MODEL FAMILIES**
Evaluate SVM, LightGBM, CatBoost, and MLP-NN under a consistent setup.
- 3 ANALYZE MODEL BEHAVIOR**
Use confusion matrices and feature importance to interpret classification decisions.

DATASET OVERVIEW

1,115,152

cleaned records

12

monthly files

5

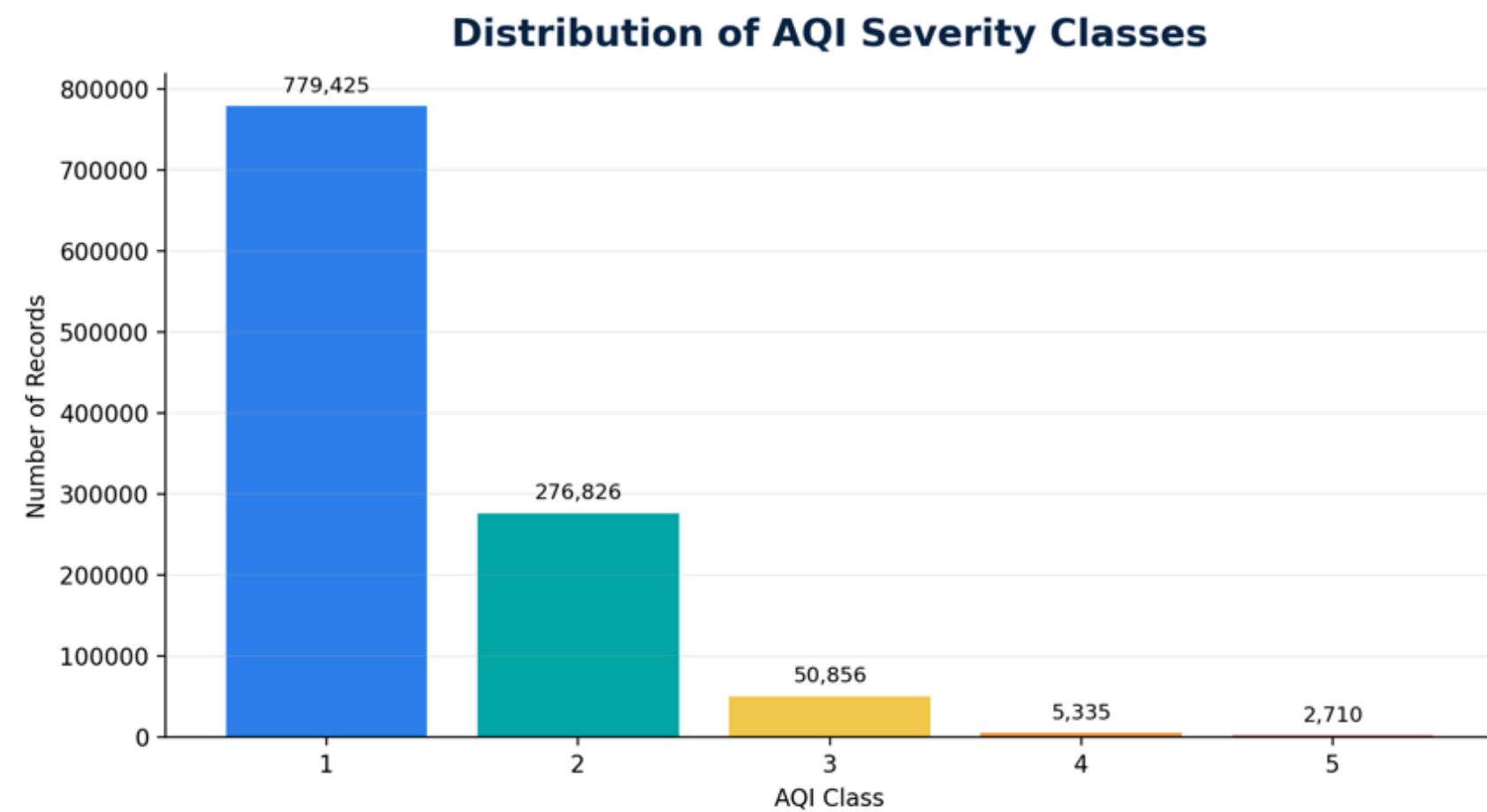
AQI classes

Target

main.aqi – ordinal AQI
severity class, **1–5**
scale

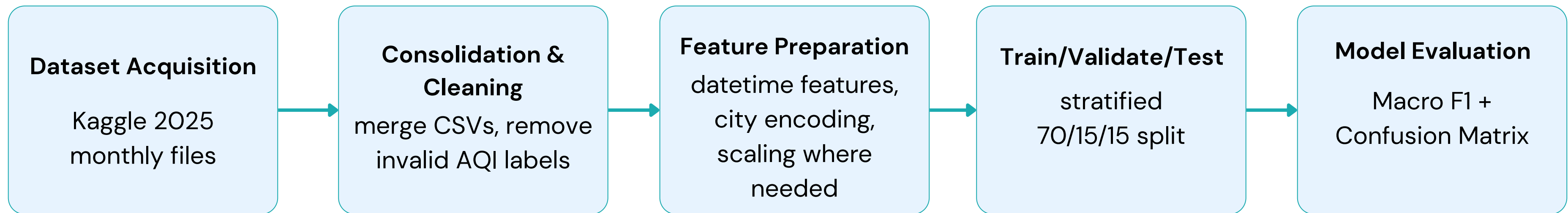
Features

pollutants + datetime
+ city_name



PH Philippine Cities Air Quality Index Data 2025 from Kaggle

METHODOLOGY PIPELINE



MODELS USED

SVM

Baseline

Margin-based classifier; linear SVM used for scalability.

CatBoost

Categorical-aware boosting

Handles city_name as categorical context.

LightGBM

Gradient Boosting

Efficient tree ensemble for structured tabular data.

MLP-NN

Neural Network

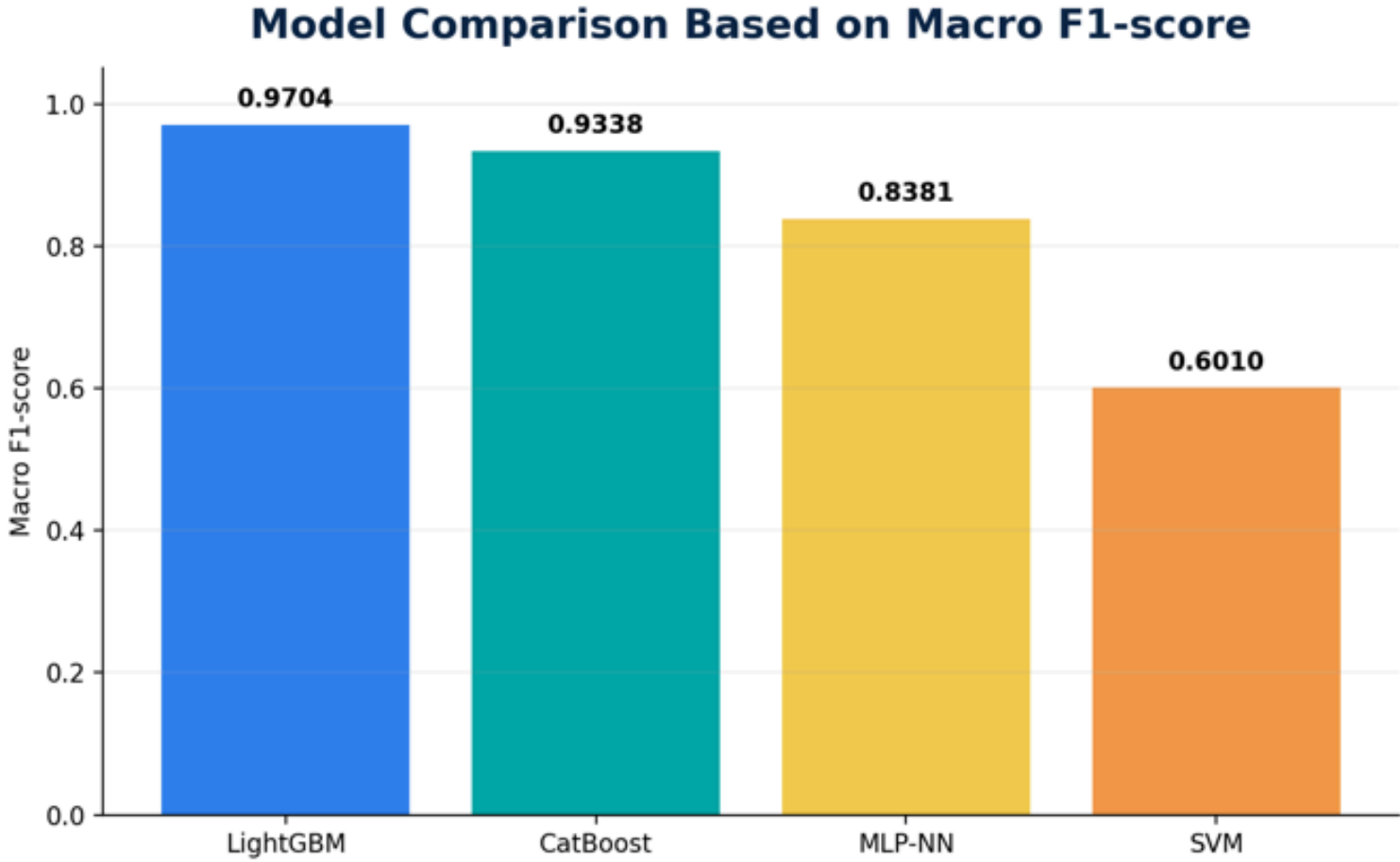
ReLU hidden layers, softmax output, Adam optimizer.

IMPLEMENTATION SETUP

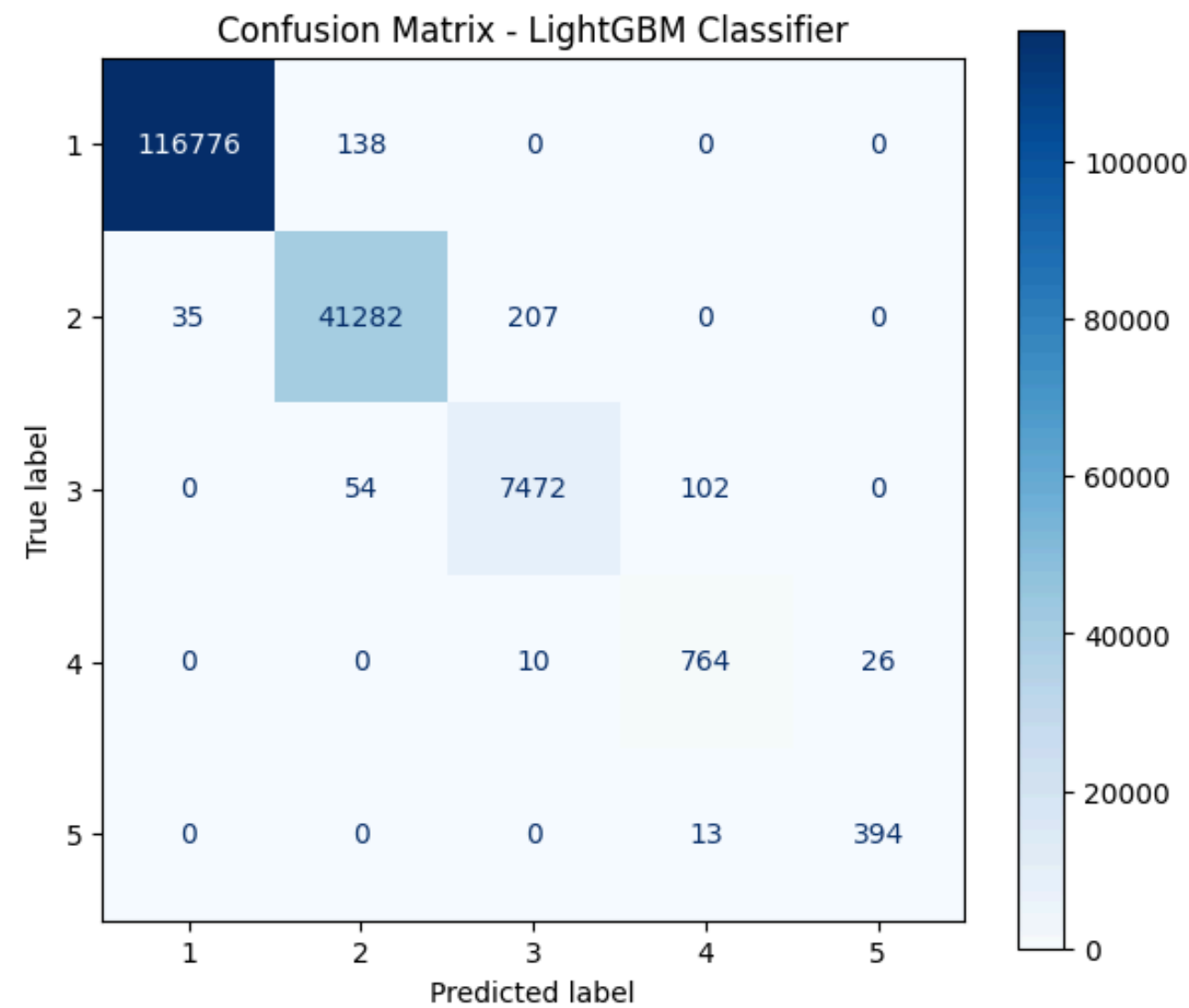
- Python implementation with pandas, NumPy, scikit-learn, LightGBM, CatBoost, and TensorFlow/Keras.
- Invalid decimal AQI labels were removed; valid classes 1–5 were retained.
- city_name was encoded as a categorical variable; numerical features were scaled for SVM and MLP-NN.
- Stratified 70/15/15 split preserved class distribution across train, validation, and test sets.
- MLP-NN used ReLU hidden layers, softmax output, Adam optimizer, and sparse categorical cross-entropy.

RESULTS: MODEL PERFORMANCE

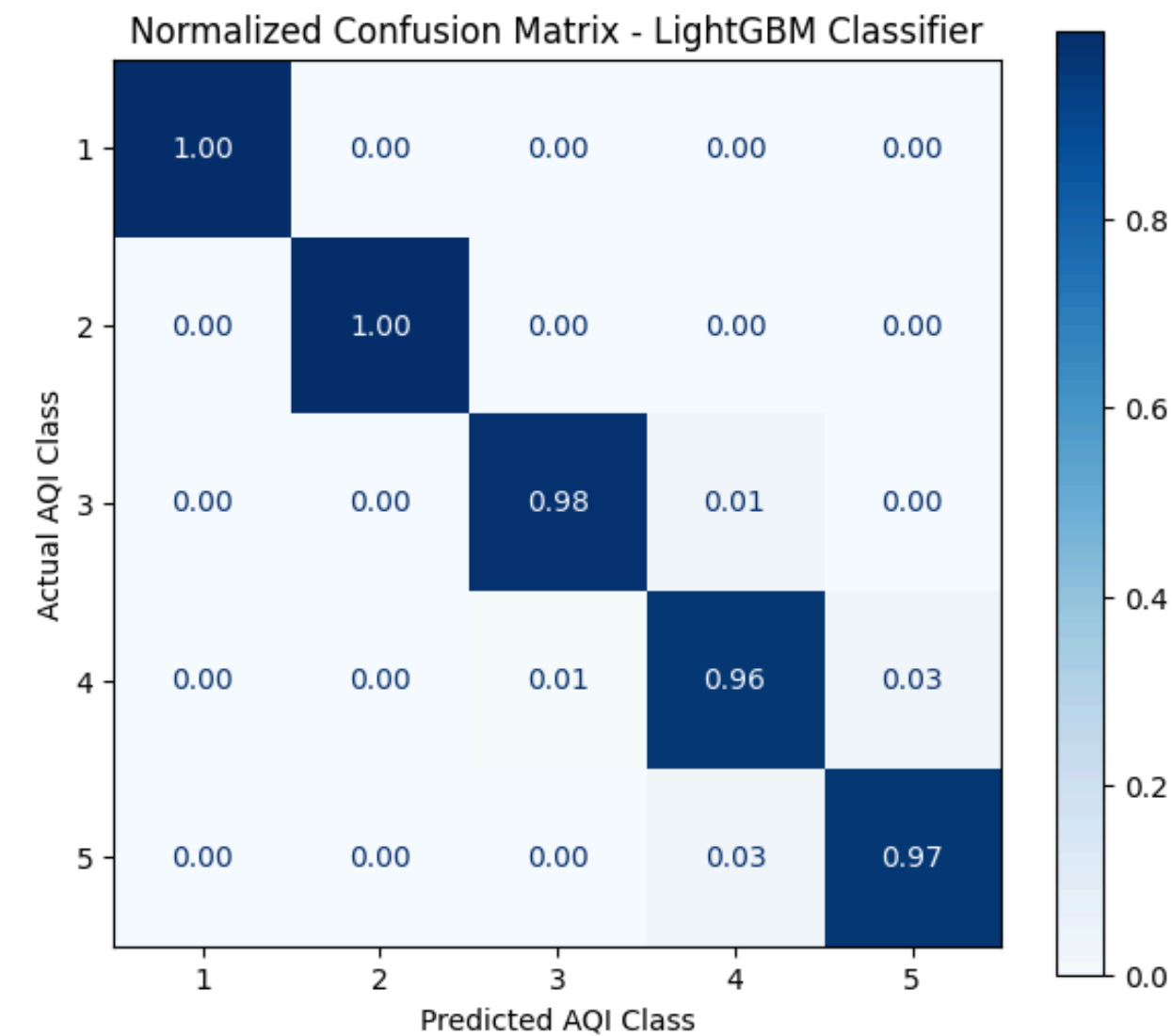
Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
LightGBM	0.9969	0.9599	0.9817	0.9704	0.9970
CatBoost	0.9926	0.9118	0.9591	0.9338	0.9927
MLP-NN	0.9493	0.8926	0.8013	0.8381	0.9489
SVM	0.8461	0.6853	0.5687	0.6010	0.8431



LIGHTGBM CLASSIFICATION ANALYSIS

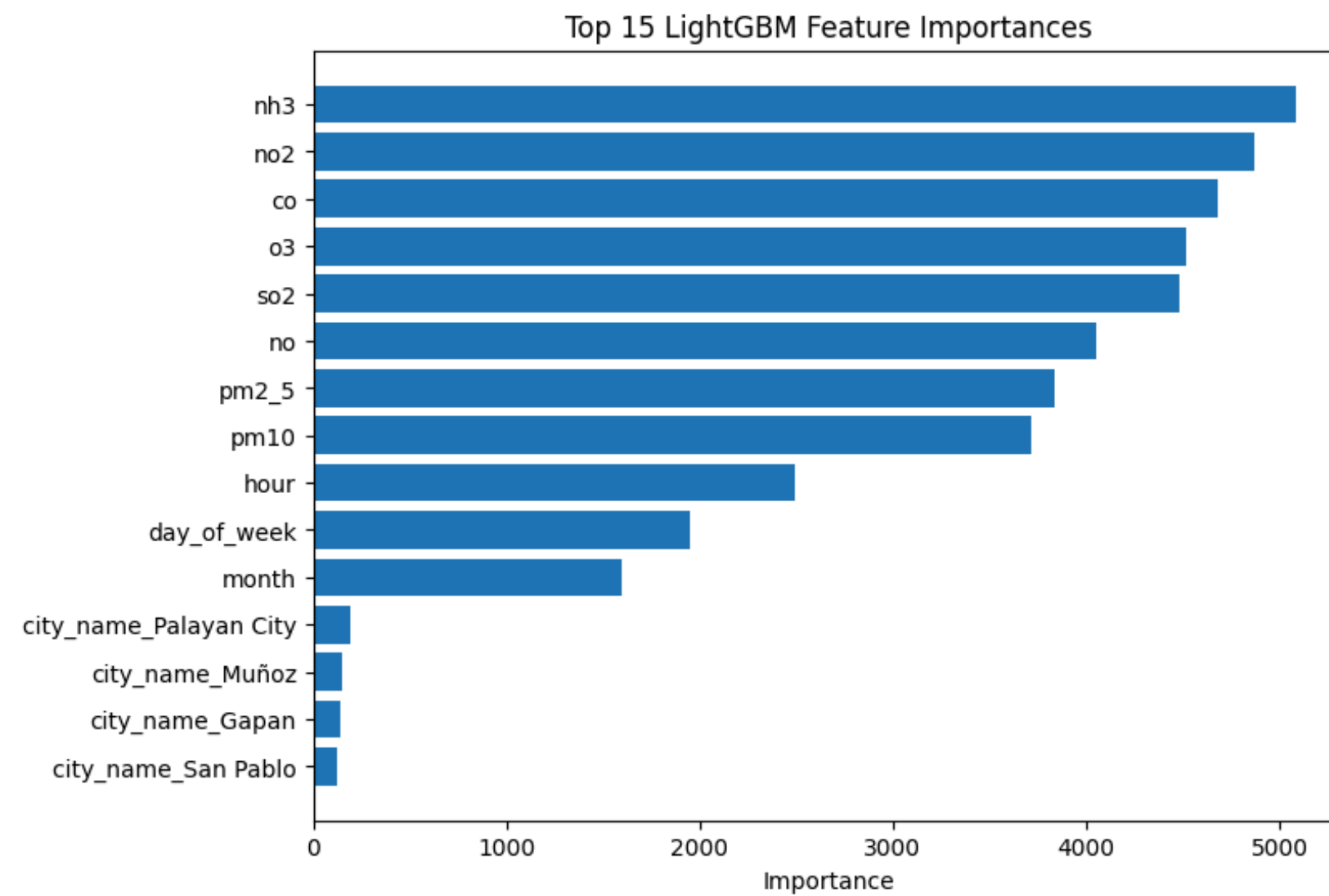


Raw Counts



Normalized Rates

FEATURE IMPORTANCE



Main Insights

- Top features were mostly pollutant components: NH_3 , NO_2 , CO, O_3 , SO_2 , NO, PM2.5, and PM10.
- Temporal features contributed, but less than core pollutant variables.
- City-based variables had lower importance, suggesting the model did not rely mainly on city identity.
- Feature importance shows predictive contribution, not direct causality.

DISCUSSION: KEY FINDINGS

- Gradient boosting models were most effective for structured AQI data.
- LightGBM achieved strong class-level performance despite severe class imbalance.
- MLP-NN trained steadily but did not outperform tree-based ensembles.
- SVM provided a useful baseline but struggled with nonlinear and minority-class patterns.
- Pollutant values were more influential than city-based context.

CONCLUSION AND FUTURE WORK

Conclusion

- The study built a reproducible AQI level classification pipeline for Philippine city-level data.
- LightGBM produced the best overall results and strongest Macro F1-score.
- Pollutant concentration variables were the main drivers of classification behavior.

Future Work

- Use continuous AQI values and official monitoring station records.
- Add meteorological variables such as humidity, wind speed, and rainfall.
- Apply SHAP or other explainability tools for deeper interpretation.
- Test time-based validation for future-period generalization.

THANK YOU

FOR LISTENING TO OUR PRESENTATION